Mount Etna

The tallest and by far the largest active volcano in Europe, with a long history of major, often spectacular eruptions

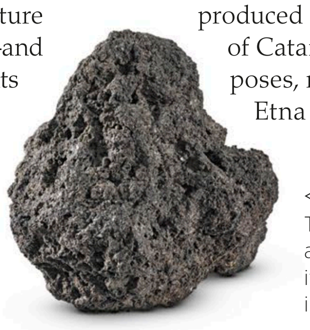
Occupying 459 square miles (1,190 square km) of eastern Sicily, Mount Etna is one of the world's largest, most famous, and most active volcanoes. Towering above Catania, it has one of the world's longest documented records of volcanic activity, dating back to 1500 BCE. Etna's origins are probably connected to it being close to the boundary between the African and Eurasian Plates; it may also lie over a mantle hotspot.

Complex structure

Etna is a stratovolcano, with a complex structure that includes four separate summit craters—and more than 300 smaller parasitic volcanic vents and cones on its flanks. Its summit is at 10,925 ft (3,330 m), and lava flows cover much of the surface of its flanks. Although Etna has an overall conical shape, there is a deep and prominent horseshoe-shaped depression on its eastern side, known as the Valle del Bove.

Eruption types

Over the past several thousand years, Etna has been almost continuously active, producing eruptions of two main types. Spectacular explosive eruptions flare up from one or more of its summit craters. These can produce tall fountains of fiery lava, volcanic bombs, cinder showers, and large ash clouds. Quieter eruptions, with large-scale outpourings of lava, also emerge from flank vents and fissures. Etna's most destructive eruption occurred in March 1669. It produced massive lava flows that destroyed most of Catania's city walls. Despite the danger it poses, most of the inhabitants of Sicily regard Etna as a major asset.



◁ LAVA LUMP
The lava that erupts from Etna is mostly of a composition and temperature that makes it flow several miles before solidifying into lumps like this.

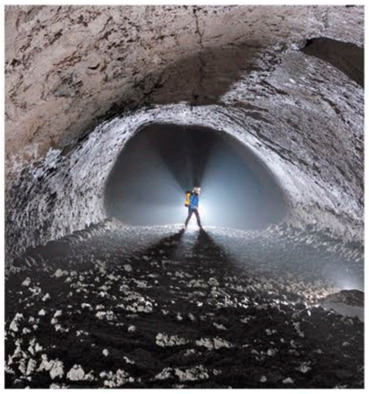
In **1999**, Etna erupted **lava fountains 1 ¼ miles (2 km) high**—the highest ever recorded

HISTORIC LAVA FLOWS

This map shows areas covered by Etna's lava flows over the centuries, going back to the 17th century. It also shows the extent of older flows. Many eruptions have come from Etna's flanks rather than its summit, and some flows have invaded urban areas. The pre-Etnian sediments are more than half a million years old—Etna's approximate age.

KEY
- △ Summit of Mount Etna
- Historic summit lavas

Flank lava flows
- 21st century
- 20th century
- 19th century
- 18th century
- 17th century
- Pre-16th century
- Prehistoric lavas
- Pre-Etnean sediments
- Urban areas



△ FIRE TUNNEL
This cave under Etna's southeastern flank is a lava tube—a natural tunnel within a solidified former lava flow. Fiery magma once flowed through it.